

The Infinite-Bagged RSM Excess Risk R_k

Exact Results

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Abstract

Under the common-loading forecast-error structure, the infinite-bagged RSM excess risk $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$ is exactly the squared distance between the bagged weight vector and the oracle weight vector, measured in the error-covariance metric. From this single identity we obtain a computable, assumption-free two-sided bracket $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$ — so the optimal subset size k^* lies between the cube and square roots of T — together with a sharp dichotomy in which bounded coherence with non-degenerate heterogeneity gives the cube root ($\alpha = 2$) while a dominant forecaster gives the square root ($\alpha = 1$), and an intermediate crossover in between. The guiding principle: which assumption is binding determines how far the ceiling of the bracket is pulled down.

Reader's summary.

1. With some heterogeneity, $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$, so the optimal k rate lies between the cube root and the square root of T .
2. Bounded coherence and non-degenerate heterogeneity $\Rightarrow R_k^{\text{bag},\infty}$ follows the power form $\asymp z_k^2 \Rightarrow k^* \asymp T^{1/3}$ (cube root).
3. High coherence with a dominant forecaster $\Rightarrow R_k^{\text{bag},\infty} \asymp z_k \Rightarrow k^* \asymp T^{1/2}$ (square root); partial coherence interpolates between the two.

1 Setup and notation

Assume the common-loading covariance of the forecast errors,

$$\Sigma_e = q \mathbf{u} \mathbf{u}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad \tau_i = \sigma_i^{-2} > 0,$$

where $q \geq 0$ is the variance of the common (unforecastable) shock and τ_i is the precision of forecaster i . Write

$$A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \quad \ell_i = \frac{\tau_i}{A_m}, \quad A_S = \sum_{j \in S} \tau_j, \quad \langle a, b \rangle_\tau = \sum_i \frac{a_i b_i}{\tau_i}.$$

RSM draws size- k subsets S uniformly, fits the constrained (sum-to-one) optimal weights $v_{S,i} = \mathbf{1}\{i \in S\} \tau_i / A_S$ within each, and bags. The infinite-bagging ($G \rightarrow \infty$) weight is $\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}]$, and the object of interest is

$$R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \bar{v}_k' \Sigma_e \bar{v}_k - Q^{\text{opt}}.$$

Two functionals carry the story, together with z_k :

$$V_\delta = \frac{1}{m} \sum_i \left(\frac{\tau_i}{\bar{\tau}} - 1 \right)^2 = \frac{\text{Var}(\tau)}{\bar{\tau}^2} \quad (\text{dispersion}), \quad H = \overline{\sigma^2} \bar{\tau} \geq 1 \quad (\text{Cauchy-Schwarz gap}), \quad z_k = \frac{1}{k} - \frac{1}{m}.$$

2 R_k is a squared weight deviation

Everything rests on one identity: $R_k^{\text{bag},\infty}$ is the squared distance of the bagged weights from the oracle weights.

Proposition 1 (Oracle weights are leverage scores). *The constrained-optimal full combination is $w_i^* = \tau_i/A_m = \ell_i$, with $Q^{\text{opt}} = q + 1/A_m$; for any subset, $Q(S) = q + 1/A_S$.*

Proof. Since $\mathbf{1}'w = 1$, the term $w'q\mathbf{1}'w = q$ is constant, so minimizing $w'\Sigma_e w$ reduces to minimizing $\sum_i w_i^2 \sigma_i^2$ subject to $\sum_i w_i = 1$; the Lagrange solution is $w_i \propto \tau_i$, giving $Q^{\text{opt}} = q + \sum_i \ell_i^2 / \tau_i = q + 1/A_m$. The subset claim is the same argument restricted to S . $\square$

Proposition 2 (Exact quadratic-form identity). *Because w^* is the constrained minimizer, the cross term vanishes and*

$$R_k^{\text{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \|\bar{v}_k - w^*\|_{\tau}^2 = \sum_i \frac{(\bar{v}_{k,i} - \ell_i)^2}{\tau_i} = \frac{1}{m^2} \sum_i \tau_i \left(g_i - \frac{1}{\bar{\tau}} \right)^2,$$

with $g_i(k) = k \mathbb{E}_S[1/A_S \mid i \in S]$ and $\bar{v}_{k,i} = \tau_i g_i(k)/m$.

Proof. $\Sigma_e w^* = (q + 1/A_m)\mathbf{1} = Q^{\text{opt}}\mathbf{1}$, and $\mathbf{1}'(\bar{v}_k - w^*) = 0$ (both sum to one), so the cross term and the common factor $q\mathbf{1}'$ both drop, leaving $\sum_i (\bar{v}_{k,i} - \ell_i)^2 / \tau_i$. The last form uses $\ell_i = \tau_i/A_m$. $\square$

Interpretation. $R_k^{\text{bag},\infty}$ is the squared weight error in the Σ_e -metric: each coordinate's error $(\bar{v}_{k,i} - \ell_i)^2$ is weighted by $1/\tau_i = \sigma_i^2$, so mis-weighting a noisy forecaster is penalized more. At infinite bagging there is no Monte-Carlo variance, so $R_k^{\text{bag},\infty}$ is the pure systematic gap of the bagged weights from optimal. The whole tuning problem trades this (decreasing in k) against estimation noise.

Proposition 3 (Endpoints: averaging $\rightarrow$ oracle). $\bar{v}_1 = \frac{1}{m}\mathbf{1}$ (simple averaging), so R_1 is the excess risk of equal weights; $\bar{v}_m = w^*$, so $R_m = 0$. Thus k traces a path from averaging to the oracle, and

$$R_1 = Q^{\text{ave}} - Q^{\text{opt}} = \frac{H - 1}{m\bar{\tau}}, \quad Q^{\text{ave}} = q + \frac{1}{m^2} \sum_i \sigma_i^2.$$

3 Exact bounds: the two-sided bracket

All statements here hold for any positive precision vector and every $1 \leq k \leq m$ — no distributional assumption, no expansion, no asymptotics.

Lemma 1 (Two exact moment identities). $\sum_i \tau_i g_i = m$ (conservation) and $\sum_i g_i = km \mathbb{E}_S[1/A_S]$.

Proposition 4 (Two-sided bracket). *For all $1 \leq k \leq m$,*

$$\underbrace{\frac{(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}}}_{\text{Gram / projection floor}} \leq R_k^{\text{bag},\infty} \leq \underbrace{\mathbb{E}_S[1/A_S] - \frac{1}{A_m}}_{\text{Jensen ceiling}}.$$

Both ends are computable from $\{\tau_i\}$ and $\mathbb{E}_S[1/A_S]$. The denominator $\sum_i \sigma_i^2 - m/\bar{\tau} = m^2(Q^{\text{ave}} - Q^{\text{opt}}) \geq 0$ (Cauchy-Schwarz), $= 0$ iff all τ_i equal. The floor is exact for ≤ 2 distinct precision types; a hierarchy of exact lower bounds follows by appending higher moments to Lemma 1.

Proof. Ceiling: w^* is the global sum-to-one minimizer (lower) and, by Jensen on the convex $v \mapsto v' \Sigma_e v$, $Q_k^{\text{bag}, \infty} \leq \mathbb{E}_S[v'_S \Sigma_e v_S] = Q_k^{\text{sub}}$ (upper); subtract Q^{opt} and use $Q(S) = q + 1/A_S$. *Floor:* minimize $\sum_i \tau_i g_i^2$ over vectors meeting the two identities of Lemma 1; the Gram minimum equals $\frac{m}{\bar{\tau}} + \frac{m^2(k\mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i 1/\tau_i - m/\bar{\tau}}$, and substituting into Proposition 2 cancels the oracle term. $\square$

Equivalent projection form. Writing $\delta_k = \bar{v}_k - w^*$ and $\eta = w^{\text{ave}} - w^*$, the identity gives $\|\delta_k\|_\tau^2 = R_k^{\text{bag}, \infty}$ and $\|\eta\|_\tau^2 = Q^{\text{ave}} - Q^{\text{opt}}$, and the floor is the Cauchy-Schwarz projection $R_k^{\text{bag}, \infty} \geq \langle \delta_k, \eta \rangle_\tau^2 / \|\eta\|_\tau^2$ with $\langle \delta_k, \eta \rangle_\tau = \frac{k}{m}(Q_k^{\text{sub}} - q) - (Q^{\text{opt}} - q)$. The floor thus uses only the three aggregate risks $Q_k^{\text{sub}}, Q^{\text{ave}}, Q^{\text{opt}}$ (and the level q); the ceiling uses only $Q_k^{\text{sub}}, Q^{\text{opt}}$ and is q -free.

Corollary 1 (Open bracket). *To leading order, with $\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$ and $\text{Var}_S(A_S) = k \frac{m-k}{m(m-1)} m\bar{\tau}^2 V_\delta$,*

$$\underbrace{\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2}_{\text{floor}} \lesssim R_k^{\text{bag}, \infty} \lesssim \underbrace{\frac{1}{\bar{\tau}} z_k}_{\text{ceiling}}.$$

Remark 1 (The two ends answer to different things). *Since $0 < z_k < 1$, $z_k^2 < z_k$: the bracket has a gap between exponents, so unconditionally $R_k^{\text{bag}, \infty} \sim z_k^\alpha$ with $\alpha \in [1, 2]$ and $k^* \in [\Theta(T^{1/3}), \Theta(T^{1/2})]$. The **floor** z_k^2 comes from the dispersion V_δ and is present whenever the forecasters are heterogeneous; the **ceiling** z_k is the Jensen bound, first order because it cannot see that bagging cancels the leading term. The effective exponent is read off the ceiling: the floor is always z_k^2 , and where α lands is decided by how far the ceiling can be pulled down.*

Proposition 5 (Monotonicity and the exact k^* condition). *If each $g_i(k)$ is monotone in k , then $R_k^{\text{bag}, \infty}$ strictly decreases in k : $\frac{dR_k^{\text{bag}, \infty}}{dk} = \frac{2}{m^2} \sum_i \tau_i (g_i - 1/\bar{\tau}) \dot{g}_i < 0$ (using $\sum_i \tau_i \dot{g}_i = 0$). Hence in population there is no bias-variance trade-off; an interior optimum exists only through estimation error. With $A = \sigma_\varepsilon^2 + Q^{\text{opt}}$ and idealized loss $L_k = (A + R_k^{\text{bag}, \infty}) \frac{T-1}{T-k}$, the discrete optimum satisfies*

$$(T - k^*) (R_{k^*} - R_{k^*+1}) = A + R_{k^*}.$$

(The unconditional bound $R_k^{\text{bag}, \infty} \leq R_1$ holds regardless, by Jensen on the harmonic mean.) As a by-product, RSM dominates equal-within-subset CSR for every k , since constrained-OLS minimizes $w' \Sigma_e w$ over the choice $1/k$ inside each subset.

4 Two regimes and the crossover

The open bracket is two-sided but its exponents differ. Two minimal, checkable assumptions close it; each governs a different end.

Assumption 1 (Bounded coherence). *There are constants $0 < c \leq C < \infty$, independent of m , with $c \leq \tau_i/\bar{\tau} \leq C$.*

The upper part (C) is “no dominant forecaster”; the lower part (c) forces $A_S \geq ck\bar{\tau}$, keeping $1/A_S$ and its reciprocal moments bounded — which makes the expansion of $\mathbb{E}_S[1/A_S]$ rigorous, not heuristic.

Assumption 2 (Non-degenerate dispersion). *There is $v > 0$, independent of m , with $V_\delta \geq v$ (genuine heterogeneity).*

Theorem 1 (Cube-root regime). *Under Assumptions 1–2 there are $0 < c_1 \leq c_2 < \infty$ (depending only on c, C, v) with $\frac{c_1}{m\bar{\tau}} z_k^2 \leq R_k^{\text{bag},\infty} \leq \frac{c_2}{m\bar{\tau}} z_k^2$ for $2 \leq k < m$. Hence $\alpha = 2$ and $k^* \asymp T^{1/3}$.*

Proof sketch. Assumption 1 controls the reciprocal expansion: with $\text{Var}_R(L_{i,R}) \leq \frac{(k-1)(m-k)}{m(m-1)(m-2)} V_\delta$ one gets $|\bar{v}_{k,i}/\ell_i - 1| \leq K_c z_k (|h_i| + V_\delta)$, so by Proposition 2 $R_k^{\text{bag},\infty} \leq \frac{K_{c,C}}{m\bar{\tau}} (V_\delta + V_\delta^2) z_k^2$. The floor of Corollary 1 is bounded below by Assumption 2. Inserting $R_k^{\text{bag},\infty} \asymp C_R z_k^2$ into Proposition 5 gives $Ak^3 + 3C_R k - 2C_R T = 0$, i.e. $k^* \approx (2C_R T/A)^{1/3}$. $\square$

Minimality. Each assumption guards a different end of the bracket:

Assumptions	Floor	Ceiling	α	k^*
1 and 2	z_k^2	z_k^2	2	$\Theta(T^{1/3})$
2 only (no 1)	z_k^2	z_k	$[1, 2]$	$[\Theta(T^{1/3}), \Theta(T^{1/2})]$
1 only (no 2)	0	z_k^2	$[0, 2]$	up to $\Theta(T^{1/3})$
neither	0	z_k	$[0, 2]$	—

Dropping bounded coherence leaves the floor at z_k^2 but springs the ceiling back to the Jensen z_k : a dominant/negligible forecaster ($\tau_{\min}/\bar{\tau} \rightarrow 0$) gives $k^* \asymp T^{1/2}$; a homogeneous panel ($V_\delta \rightarrow 0$) gives $R_k^{\text{bag},\infty} \rightarrow 0$ and averaging is optimal. So the cube root is exactly the bounded-coherence, heterogeneous regime.

Remark 2 (The intermediate case is a crossover, not a new power). *Partial coherence does not produce a single power z_k^α with fixed $\alpha \in (1, 2)$. It is an additive crossover,*

$$R_k^{\text{bag},\infty} \approx \underbrace{\frac{\rho}{\bar{\tau}} z_k}_{\text{coherence leakage}} + \underbrace{b z_k^2}_{\text{dispersion}}, \quad \rho \in [0, 1], \quad b > 0,$$

with $\rho = 0$ the cube root and $\rho = 1$ the square root. The local slope slides from ≈ 2 at small k (where z_k^2 dominates) to ≈ 1 at large k (where z_k dominates), crossing near $k_\times \sim b\bar{\tau}/\rho$. A finite-range log–log fit then reads an effective $\alpha \in (1, 2)$ — the average local slope — and k^* interpolates as $T^{1/(\alpha+1)}$, sitting near $T^{1/3}$ for moderate T and drifting toward $T^{1/2}$ as T grows whenever $\rho > 0$.

Remark 3 (What is exact and what needs a form). *The intermediate region ($k^* \in [T^{1/3}, T^{1/2}]$) is exact — it is the bracket, no functional form assumed; likewise a computed k^* for a given profile is obtained by minimizing the exact L_k over integer k . Only a clean single exponent or closed-form scaling needs the power form of Remark 2, and that form is anchored ($R_m = 0$, $R_1 = (H-1)/(m\bar{\tau})$, and the leading curvature) and falsifiable against computed $R_k^{\text{bag},\infty}$, not free. Two qualifiers: the bracket itself is exact even at fixed m (only the uniform-in- m rate needs Assumptions 1–2 with m -free constants); and everything here is the population object — the map to a realized finite- G , finite- T MSFE adds the inflation factor $\frac{T-1}{T-k}$ and a Monte-Carlo ensemble term.*

5 Interpretation

One dial, two costs. By Proposition 2, $R_k^{\text{bag},\infty}$ is the squared distance of the bagged weights from the oracle weights, and it decreases in k (larger subsets capture more of the covariance structure). Estimating the within-subset weights from T observations costs a variance that increases in k and decreases in T . The subset size is therefore a single bias–variance dial: $k = 1$ is pure averaging (no estimation, robust, biased if heterogeneous), $k = m$ is full constrained OLS (no bias, maximal estimation noise), and the optimum $k^* \propto T^{1/(\alpha+1)}$ balances the two. It rises when there is more structure to capture and when more data T are available to capture it.

When equal weights are already good. If the forecasters are homogeneous ($V_\delta \approx 0$, $H \approx 1$), then $R_1 = Q^{\text{ave}} - Q^{\text{opt}} \approx 0$: equal weights are already near-optimal and the bias is negligible at every k . One keeps k *low* — staying near simple averaging — and needs little T ; not because the weights are easy to estimate, but because there is essentially nothing to estimate. The gain from optimal combination is small (the regime in which averaging wins).

When a few forecasts dominate. If some forecasters are far more precise than the rest, the oracle weights are far from equal (R_1 and H large), so there is real bias to remove and a real gain to capture. One pushes k *up*, because only larger subsets up-weight the precise forecasters, and one needs more T , because the large, differentiated weights must now be estimated from data. In the extreme this is the square-root regime, $k^* \asymp T^{1/2}$ — larger, and growing faster in T , than the cube root.

The mechanism: how fast the weights converge. The deepest reading uses $R_k^{\text{bag}, \infty} = \|\bar{v}_k - w^*\|_\tau^2$. Under bounded coherence every weight coordinate converges to its target at a uniform, fast rate as k grows, so the gap closes like z_k^2 and a modest k suffices (cube root). With a dominant forecaster, that coordinate is heavily under-weighted at equal weights and converges *slowly* — it must be “found” through random subsets and given proper weight only as k grows — so the gap closes only like z_k and a large k is needed (square root). The bagging exponent is the convergence rate of the slowest weight.

One good forecaster, or several? What matters is not the number of good forecasters but their *share of the total precision*. If g forecasters carry essentially all the precision, the coherence is $\kappa = \tau_{\max}/\bar{\tau} \approx m/g$. A vanishing fraction ($g \ll m$, a small clique amid many noisy forecasters — the single dominant forecaster being the extreme $g = 1$) makes κ large, breaking bounded coherence: square root, large k^* , more T . A substantial fraction ($g \propto m$) keeps κ bounded: cube root. This is the common-loading face of the Elliott–Liao result that the worst case for averaging is to average over a good subset of size m_1 and discard the rest, with the gain large precisely when m_1 is small relative to m . “Dominant” means dominant in *share of precision*, whether that is one forecaster or a few.